

Question 1

- a) IOI City Mall has recently introduced an automated *License Plate Recognition (LPR)* system for its car parking at all levels. The system allows shoppers to drive through upon entry and exit the parking without the issuance of printed parking tickets as it will capture vehicle plate numbers using the *LPR* camera installed at the parking entry and exit points.
- (i) *PEAS*, which stands for *Performance, Environment, Actuator* and *Sensor*, is an effective way of describing how an intelligent system works. Construct a *PEAS* description for the problem to which the *LPR* is going to solve. (4 marks)

PEAS	Description
Performance (P)	Accuracy of recognizing car plate number
Environment (E)	Car park, cars, humans, animals
Actuator (A)	Display screen (to display the scanned result)
Sensor (S)	Camera, pressure/weight sensor

- (ii) Based on information provided below, identify **SIX (6)** properties of *LPR*'s environment. Justify your answer. (6 + 6 marks)
- Fully observable, partially observable or non-observable
 - Episodic or sequential
 - Single-agent or multi-agent
 - Static or dynamic
 - Deterministic or non-deterministic
 - Discrete or continuous (in terms of percept)

Property	Justification
Fully observable	Due to the predictable position of a car plate on every car, the agent can always "see" the entire car plate.
Single agent	The agent is solving car plate recognition problem by itself.

Non-deterministic	When recognition is performed twice, different outcomes might be produced, and this depends on the presence of obstacles (e.g., walking humans)
Episodic	The result of recognizing one car plate has no impact on the result of recognizing the following car plate.
Dynamic	When the agent is screening the car plate for percept, the environment is changing.
Discrete	Car will definitely stop due to the presence of barrier. So, regardless of whether a camera or video recorder is used, there will be a finite number of frames which will be used as percept.

- b) *ChatGPT* is a different type of intelligent system from the *LPR*. It is an *Artificial Intelligence (AI)* chat-bot that can create human-like conversational dialogue. Two months right after its first launch at the end of year 2022, it already has more than 100 million active users. Its achievement has surpassed many users' expectations. One important reason for its popularity is its learning capability.

- (i) A learning agent usually consists of four *conceptual components*. Identify the **FOUR (4)** *conceptual components*. Then, briefly describe each of them. (4 + 4 marks)

Component	Description
Performance element	Responsible for selecting external actions.
Learning element	Responsible for making improvements
Critic	Responsible for evaluating the agent's behavior with respect to a fixed performance standard
Problem generator	Responsible for suggesting actions that will lead to new experiences.

- (ii) Name **ONE (1)** person who is credited to the method of creating intelligent machines, that is, first build learning machines and then teach them later. (1 mark)

- Alan Turing

Question 3

- a) Suppose $P_i = \{P_1, P_2, P_3, P_4\}$ represents four participants of a *Creative Content Competition* who are nominated for the *Best Creative Idea Award*. The domain of each variable is the set $D_i = \{0, 1\}$ where 0 and 1 mean “not awarded” and “awarded” respectively. For instance, when $P_i = 1$, it means that P_i will receive the award; and when $P_i = 0$, it means that P_i will not receive the award. For each of the following statement, write a constraint in **TWO (2)** different ways: 1) an *abstract relation*, and 2) an *enumeration of the members of the relation*; and then indicate whether it is a *unary constraint*, a *binary constraint*, or a *ternary constraint*.

- (i) Only **ONE (1)** participant will be awarded.

(3 marks)

$\langle (P_1, P_2, P_3, P_4), P_1 + P_2 + P_3 + P_4 = 1 \rangle$

$\langle (P_1, P_2, P_3, P_4), [(0,0,0,1), (0,0,1,0), (0,1,0,0), (1,0,0,0)] \rangle$

Ternary constraint

- (ii) At least **TWO (2)** participants between P_1, P_2 and P_3 will be awarded.

(3 marks)

$\langle (P_1, P_2, P_3), P_1 + P_2 + P_3 \geq 2 \rangle$

$\langle (P_1, P_2, P_3), [(0,1,1), (1,0,1), (1,1,0), (1,1,1)] \rangle$

Ternary constraint

- (iii) Only **TWO (2)** participants will be awarded.

(3 marks)

$\langle (P_1, P_2, P_3, P_4), P_1 + P_2 + P_3 + P_4 \leq 2 \rangle$

$\langle (P_1, P_2, P_3, P_4), [(0,0,1,1), (0,1,0,1), (1,0,0,1), (1,0,1,0), (1,1,0,0), (0,1,1,0)] \rangle$

Ternary constraint

- (iv) P_2, P_3 and P_4 will be awarded or will not be awarded at the same time.

(3 marks)

$\langle (P_2, P_3, P_4), P_2 = P_3 = P_4 \rangle$

$\langle (P_2, P_3, P_4), [(0,0,0), (1,1,1)] \rangle$

Ternary constraint

- b) Sudoku is a puzzle game based on the logical placement of numbers which has introduced millions of people to *Constraint Satisfaction Problem (CSP)*. **Figure 2** shows an example of 4×4 Sudoku consists of 16 squares, some of which are initially filled with digits between 1 and 4. The objective of solving a Sudoku is to fill the 4×4 grid with digits so that each column, each row, and each of the four 2×2 sub-grids that compose the grid contain digits between 1 and 4. Answer the following two questions based on the Sudoku provided in **Figure 2**.

	1	2	3	4
A		3	1	2
B	2			4
C	3			1
D			4	

Figure 2. Sudoku

- (i) Formulate the Sudoku as a *CSP* by identifying its variables, domains for each variable and constraints using the *Alldiff* global constraint. (5 marks)

Variable, $X = \{A1, \dots A4, B1, \dots B4, C1, \dots C4, D1, \dots D4\}$

Domain of empty squares, $D_i = \{1, 2, 3, 4\}$.

$C =$

row

$\text{Alldiff}(A1, A2, A3, A4)$

$\text{Alldiff}(B1, B2, B3, B4)$

$\text{Alldiff}(C1, C2, C3, C4)$

$\text{Alldiff}(D1, D2, D3, D4)$

subgrids

$\text{Alldiff}(A1, A2, B1, B2)$

$\text{Alldiff}(A3, A4, B3, B4)$

Alldiff (C1, C2, D1, D2)

Alldiff (C3, C4, D3, D4)

Column

Alldiff (A1, B1, C1, D1)

Alldiff (A2, B2, C2, D2)

Alldiff (A3, B3, C3, D3)

Alldiff (A4, B4, C4, D4)

- (ii) Suppose the *Alldiff* constraints have been expanded into binary constraints so that the *AC-3 algorithm* can be applied directly, explain how **TWO (2)** variables, *C3* and *B2*, are made arc-consistent step-by-step. It is important to note that all steps applied in making *C3* arc-consistent will remain effective when it is *B2*'s turn to be made arc-consistent.

(4 + 4 marks)

C3

- From the constraints in the box/ sub-grid, we can remove 1, 2, and 4 from B3's domain. As a result, the domain is reduced to {3}.
- From the constraints in the column, we can remove 1, 3, and 4 from C3's domain. As a result, the domain is reduced to {2}.

B2

- From the constraints in the row, we can remove 1, 2, and 3 from A1's domain. As a result, the domain is reduced to {4}.
- From the constraints in the box, we can remove 2, 3, and 4 from B2's domain. As a result, the domain is reduced to {1}.